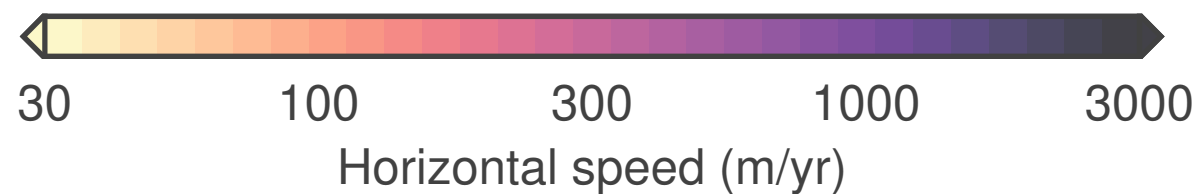


# Course syllabus

Physical principles of remote sensing  
(Ge 158)  
Fall 2026  
Brent Minchew



contains data from: Minchew et al., 2017, Gardner et al., 2018,  
Mouginot et al., 2017, and Fretwell et al., 2013

# Course description

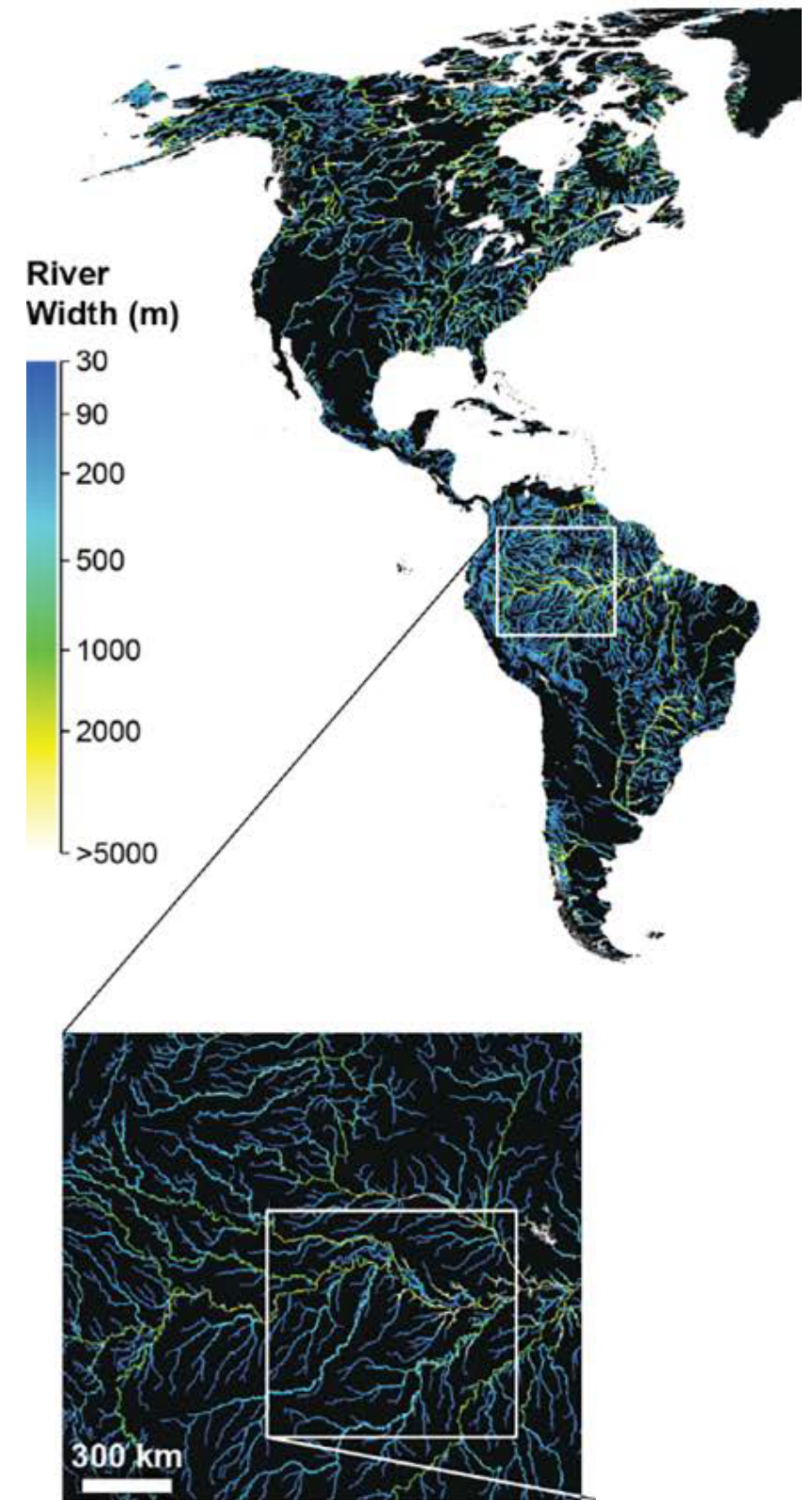
- Survey course on remote sensing, with a focus on applications in Earth and planetary sciences
- In general, we will focus on fundamental physical principles and representative applications
- Primary areas of interest:
  - Review:
    - material properties
    - electromagnetic waves
  - EM Wave propagation and attenuation
  - Interaction of EM waves with matter
  - Radiation
  - Radar



**Calving front of Thwaites Glacier, West Antarctica**  
Credit: NASA/OIB/Jeremy Harbeck

# Course objectives

- Primary: To provide students with a broad understanding of the physical principles of remote sensing that will enable them to
  - Develop strong physical intuition for problems in remote sensing
  - Think creatively about how to use remote sensing data and tools to solve problems
  - Think critically about published remote sensing data and results
  - Become developers or informed users of remote sensing data and tools
- Secondary: To provide a brief introduction to available tools and data sources that can be applied to specific problems in science and engineering

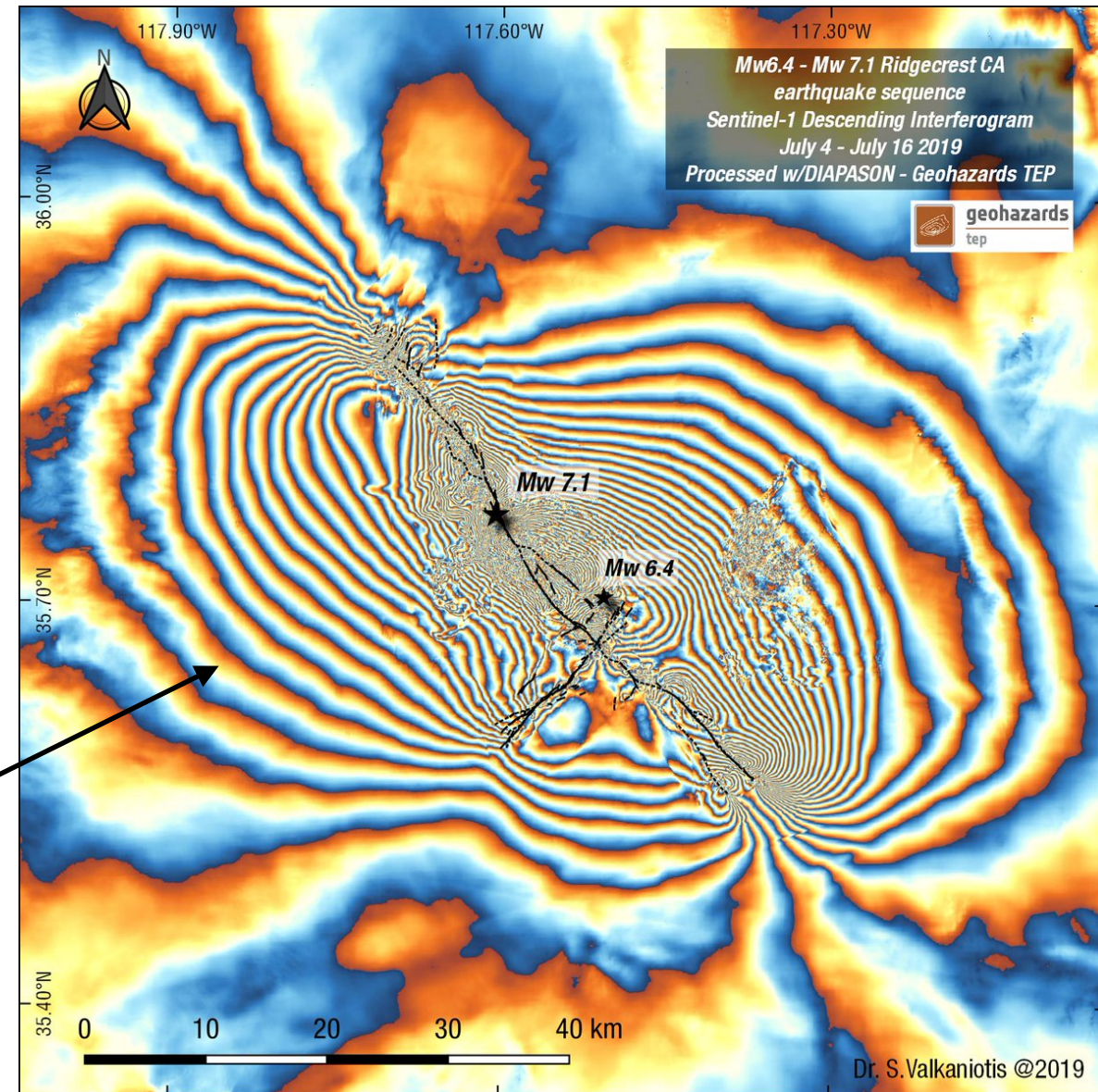


“Global extent of rivers and streams”,  
Allen and Pavelsky, 2018

# Topics to be covered during the term

- Electromagnetic (EM) spectrum and waves
- Interaction of EM waves with matter
  - Surfaces, flat and rough
  - Multiple layers
  - Volumes (vegetation, snow, atmosphere)
- Radiation
- Remote sensing systems
  - Radar (both passive and active)
- Geodesy and kinematic methods
- Selected applications

Interferogram of a recent earthquake near Ridgecrest, CA



# Course details

---

- Undergraduate and graduate students
- Prereq: introduction to electromagnetism strongly desired
- Textbook: None required. You will get everything you need for the class, but several useful references are listed in the syllabus for those who want additional resources.
- Project-based class. No exams! Grading scheme:
  - Homework: 4 psets x 10 % each = 40 % of final grade
  - Project abstract (due Sept 30) = 10 % of final grade
  - Project proposal and brief presentation (due Nov. 2) = 20 % of final grade
  - Final project report and presentation (due Dec. 5, with presentations given Dec 5 and 7) = 30 % of final grade
- 5 discussions organized around case studies of different remote sensing applications

# A bit about me

- Research ice sheet dynamics
- Primary tools:
  - Kinematic remote sensing
  - Continuum models (theoretical and numerical)
- Prof at Caltech (2025-)
- Prof at MIT (2018-2025)
- Postdoc at the British Antarctic Survey, Cambridge, UK (2016-2018)
- PhD in Geophysics at Caltech (2016)
- BS and MS in Aerospace Engineering at UT Austin (2008 and 2010)
- US Marine Corps (1996-2004)
- Born and raised in Texas

